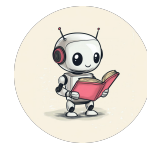




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Part 1: Business Requirement

Business Context & Goal



What's the problem ?

Santander wants to recommend relevant financial products to existing customers to increase engagement and satisfaction. Currently, recommendations are uneven: some customers receive many, others too few.

Objective



Predict the **top-K new products** each customer is most likely to adopt in the upcoming month.



BUSINESS MODEL

3 competitive advantages that make us unique



1	Customer focus	We provide a multichannel offer of products and services to fulfil all the financial needs of our customers
2	Scale	Our global and in-market scale further improves our local banks profitability by adding value and network benefits
3	Diversification	By customers: We serve a wide range of customers: individuals, SMEs, large corporates, Wealth Management customers, first-time banking customers, auto customers and dealers, and card customers By country: Our diversified geographical footprint is well balanced between developing and mature markets By balance sheet: Santander has a strong, simple and diversified balance sheet, which is highly collateralized and made up mainly of loans, with very low exposure to market risk and funding backed by retail deposits

Data Overview



1. A dataset which consists of **1.5year** customer behavior data. Approximately there are 1 million customers and the timeline for the data is from **January 2015 to May 2016**.
2. There are approximately **13.5 million** records which gives us the information about the customer and the products purchased he/she purchased.
3. There are **24 variables** which gives us the information about the customer profile, along with this there are **24 flags** which gives the information about the various products offered by the bank.
4. If a customer owns a particular product, it will be marked as 1 otherwise it will be marked as 0.

VARIABLE	VARIABLE	DATA TYPE
0 date		object
2 customer_index		int32
4 country of residence		int32
5 gender		int32
4 gender		float32
6 registration date		float16
10 new customer		float32 ²
15 last primary date		channel
16 pransion type		channel
17 relatomer type		incob ²
28 customer_index		province_name
27 addines type		income_index
23 activity_index		
24 segment		



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Part 2: Data Preparation

Data summary

Customer Features



- **Date:** Snapshot date of each customer's state (monthly partition)
- **Customer_id:** Unique customer identifier
- **Employee_index:** Customer's employment relation to the bank (e.g., employee, ex-employee)
- **Country_of_residence :** Country where the customer resides
- **Gender:** Customer's gender (e.g., female/male)
- **Age:** Age of the customer
- **Registration_date :** Date when customer first registered with the bank
- **New_customer :** Flag indicating if the customer is new (joined within last 6 months)
- **Seniority:** Customer's tenure with the bank (in months)
- **Primary_customer :** Indicates if the customer is the primary account holder
- **Last_primary_date :** Date when the customer was last primary holder (if applicable)
- **Customer_type :** Classification of customer type (e.g., primary, secondary)
- **Relation_type :** Customer relationship status at snapshot (e.g., active, inactive)

Data Summary

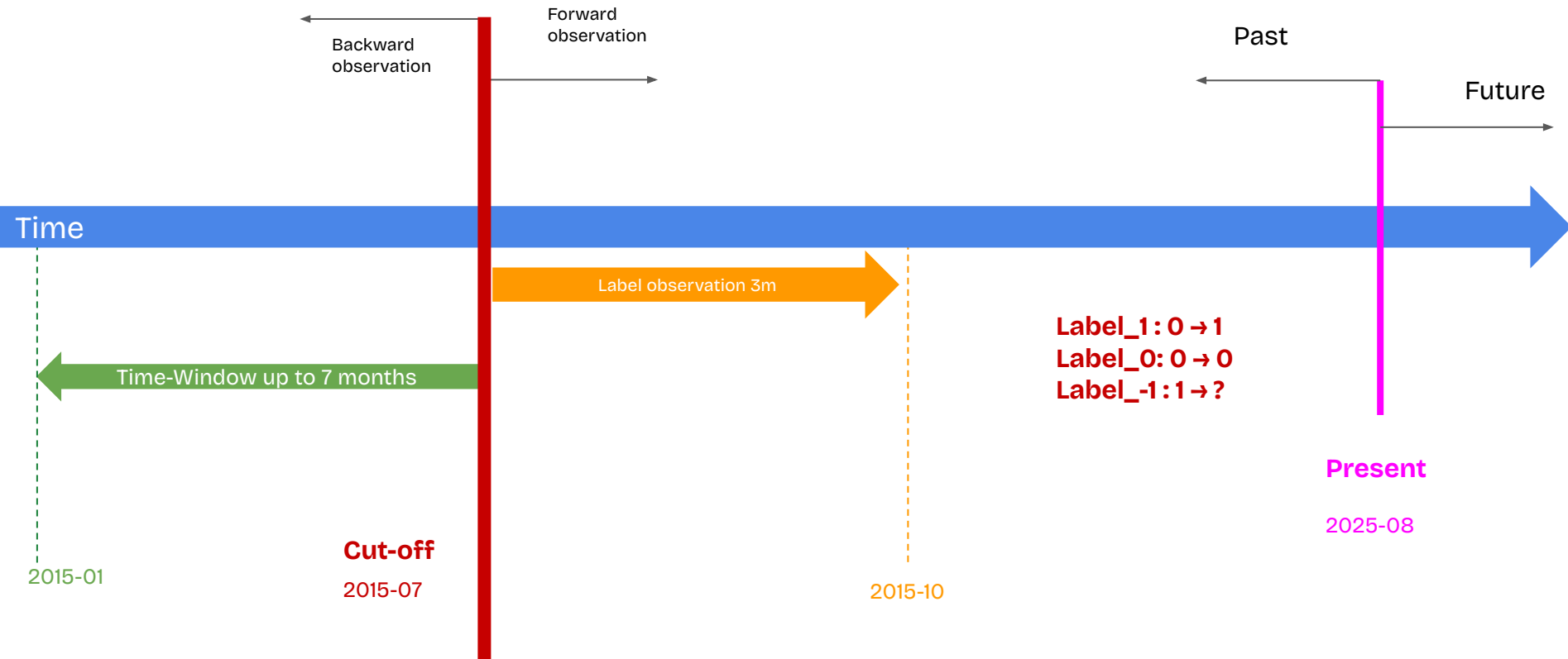
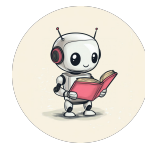
Customer Features



- **Residence_index** : Flag if customer resides in the bank's operating country
- **Foreigner_index** : Flag for foreign customer status
- **Spouse_index** : Whether customer is spouse of an employee
- **Channel** : Channel of engagement/registration (e.g., online, branch)
- **Deceased_index** : Indicates if the customer is deceased
- **Address_type** : Type of address (e.g., primary residence)
- **Province_code**: Geographical region code
- **Province_name** : Name of the region or province
- **Activity_index** : Indicator of customer activity (e.g., active/inactive)
- **Income**: Estimated or reported household income
- **Segment**: Customer segment (e.g., VIP, Personal, Graduate)

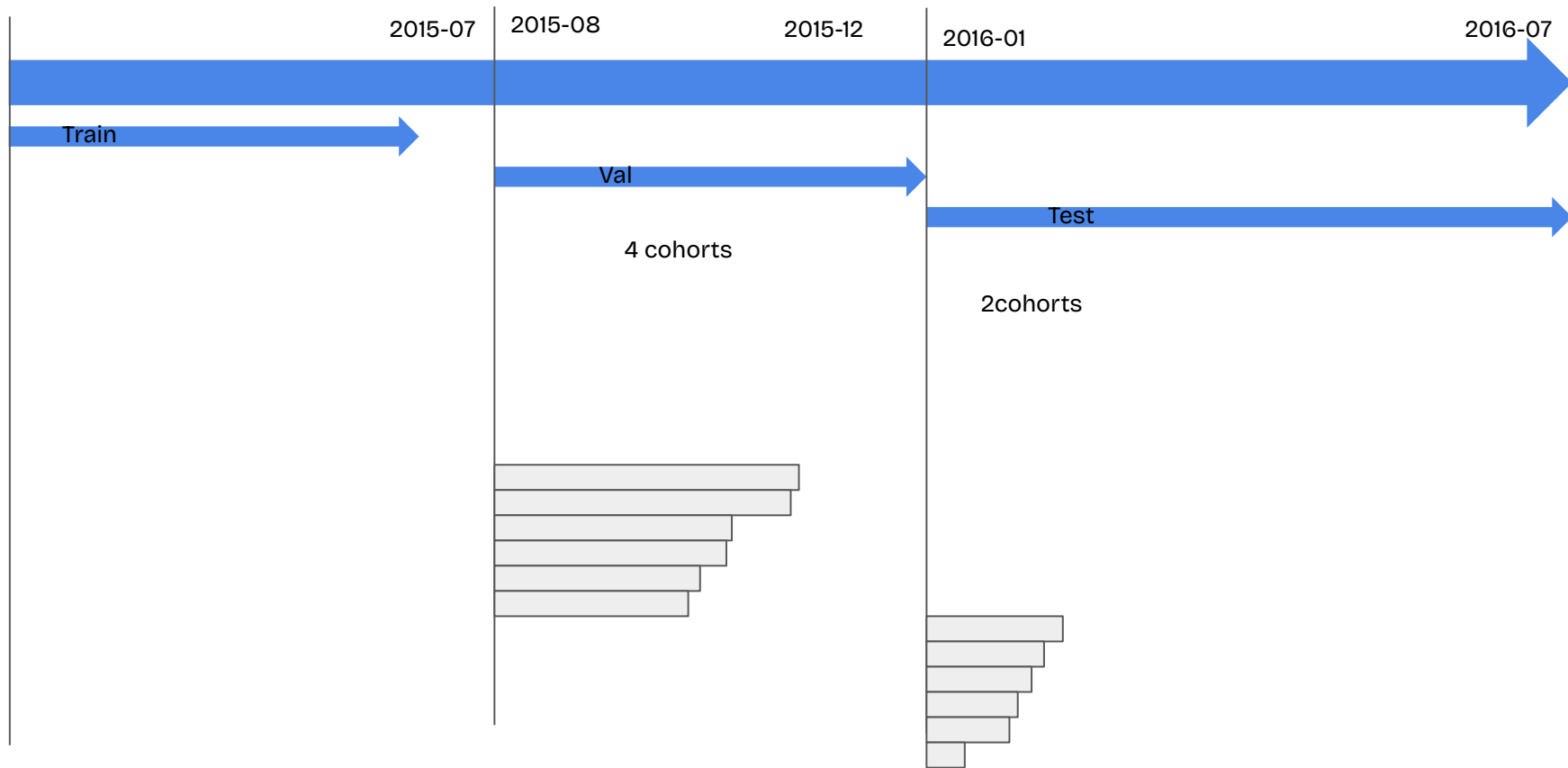
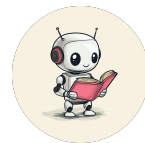
Label Definition - Look forward to next 3 months

Time-based cohort at cutoff 2015-07

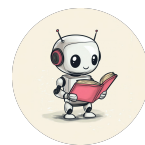


Time-based Train/Val/Test Split

Size of user's snapshot across cohorts



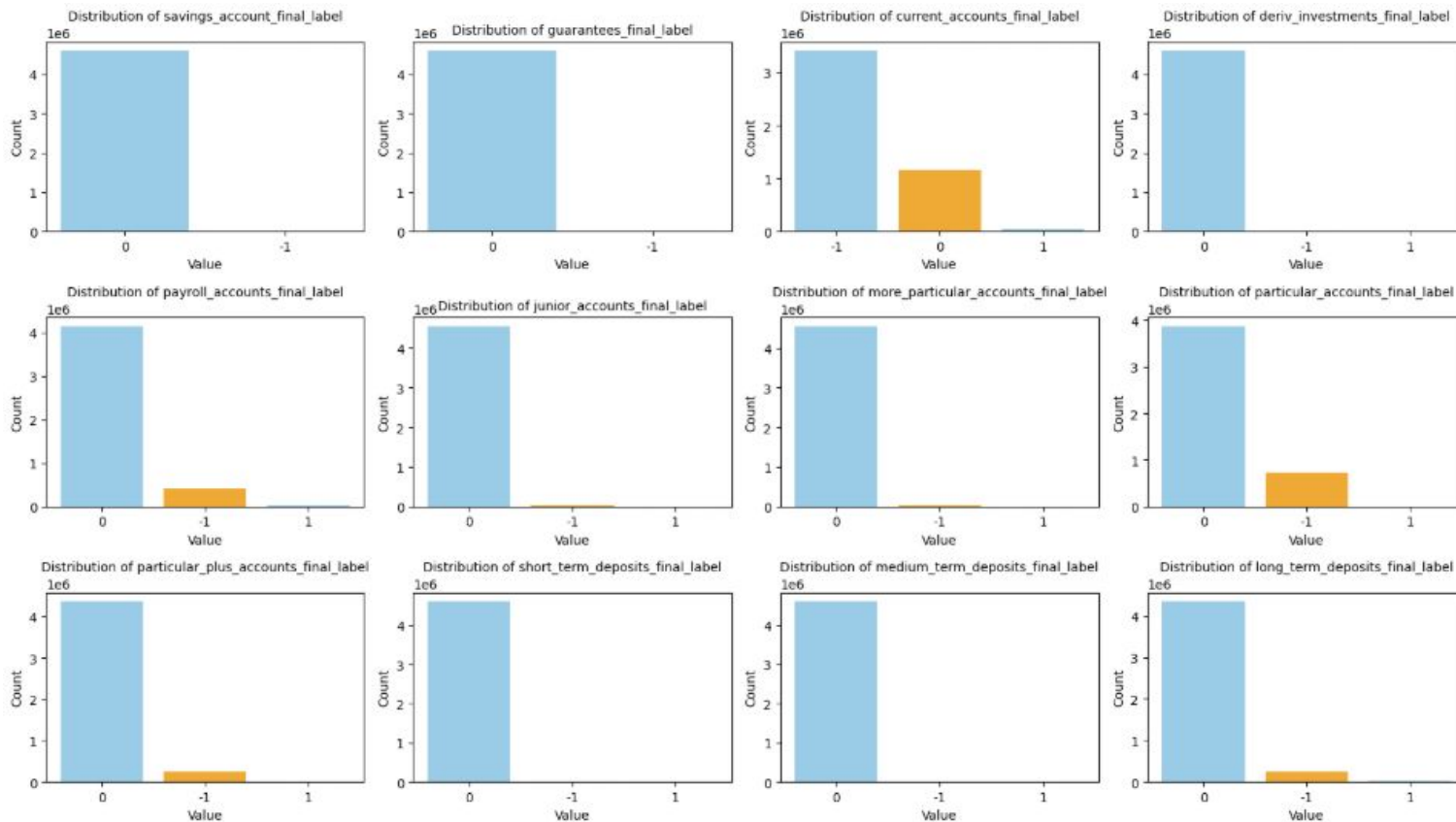
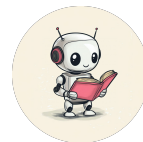
Data Quality



	Column Name	Data Type	Missing Values	% Missing	Unique Values	Outliers							
0	date	datetime64[ns]	0	0.000000	7	NaN	21	savings_account_final_label	int8	0	0.000000	2	574.0
1	customer_id	int32	0	0.000000	837157	0.0	22	guarantees_final_label	int8	0	0.000000	2	133.0
2	employee_index	object	27734	0.602087	5	NaN	23	current_accounts_final_label	int8	0	0.000000	3	0.0
3	country_of_residence	object	27734	0.602087	116	NaN	24	deriv_investments_final_label	int8	0	0.000000	3	2341.0
4	gender	object	27754	0.602521	2	NaN	25	payroll_accounts_final_label	int8	0	0.000000	3	456981.0
5	age	float64	27734	0.602087	117	30576.0	26	junior_accounts_final_label	int8	0	0.000000	3	52037.0
6	customer_tenure_days	float64	27734	0.602087	7475	0.0	27	more_particular_accounts_final_label	int8	0	0.000000	3	48687.0
7	new_customer	float16	27734	0.602087	2	77351.0	28	particular_accounts_final_label	int8	0	0.000000	3	732735.0
8	seniority	float64	27748	0.602391	247	0.0	29	particular_plus_accounts_final_label	int8	0	0.000000	3	247785.0
9	primary_customer	float16	27734	0.602087	2	6146.0	30	short_term_deposits_final_label	int8	0	0.000000	3	7509.0
10	customer_type	object	44139	0.958229	13	NaN	31	medium_term_deposits_final_label	int8	0	0.000000	3	10331.0
11	relation_type	object	44139	0.958229	4	NaN	32	long_term_deposits_final_label	int8	0	0.000000	3	260656.0
12	residence_index	object	27734	0.602087	2	NaN	33	e_account_final_label	int8	0	0.000000	3	463794.0
13	foreigner_index	object	27734	0.602087	2	NaN	34	funds_final_label	int8	0	0.000000	3	107705.0
14	spouse_index	object	4605663	99.985932	2	NaN	35	mortgage_final_label	int8	0	0.000000	3	33054.0
15	channel	object	47028	1.020947	159	NaN	36	pensions_final_label	int8	0	0.000000	3	51678.0
16	deceased_index	object	27734	0.602087	2	NaN	37	loans_final_label	int8	0	0.000000	3	14388.0
17	province_name	object	53772	1.167355	52	NaN	38	taxes_final_label	int8	0	0.000000	3	298597.0
18	activity_index	float16	27734	0.602087	2	0.0	39	credit_card_final_label	int8	0	0.000000	3	319983.0
19	income	float32	804566	17.466602	504623	228250.0	40	securities_final_label	int8	0	0.000000	3	148331.0
20	segment	object	47338	1.027677	3	NaN	41	home_account_final_label	int8	0	0.000000	3	21917.0
							42	payroll_final_label	int8	0	0.000000	3	354296.0
							43	pensions_2_final_label	int8	0	0.000000	3	386476.0
							44	direct_debit_final_label	int8	0	0.000000	3	813622.0
							45	was_primary_customer	int64	0	0.000000	2	6146.0

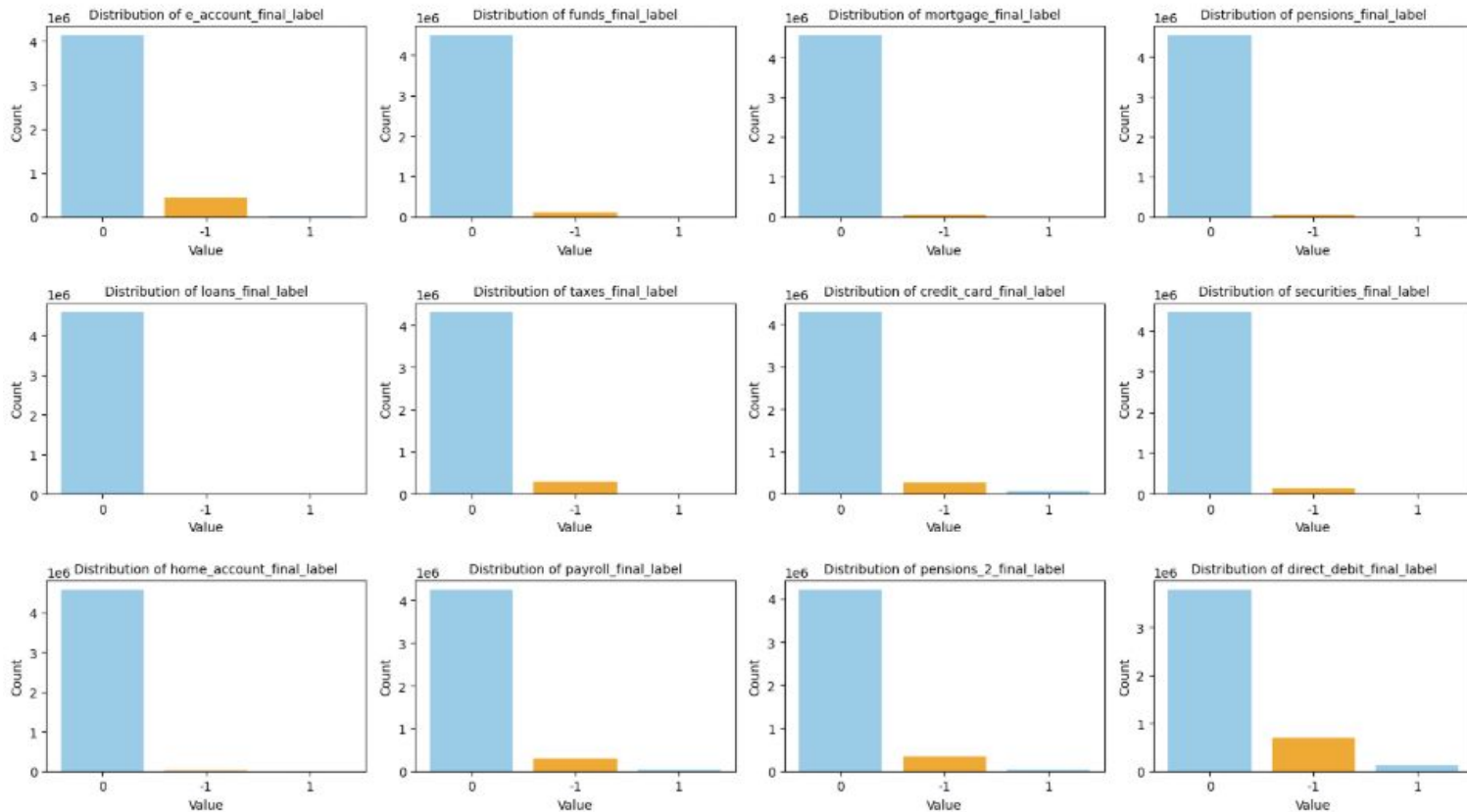
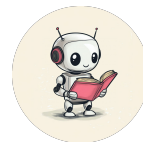
Data Exploration - Univariate Analysis

Target

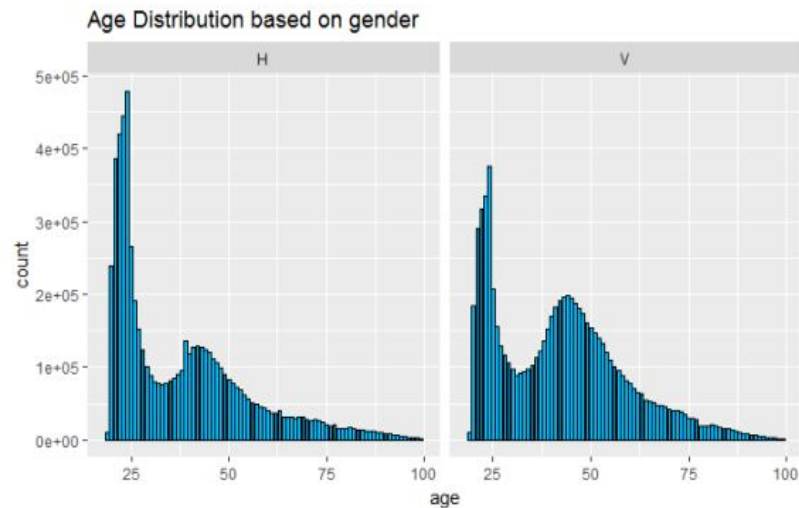
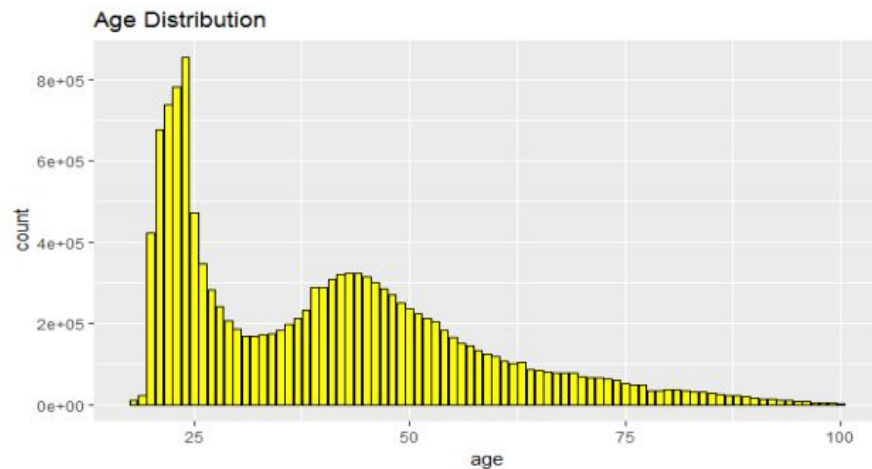
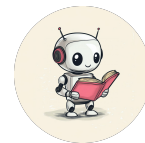


Data Exploration - Univariate Analysis

Target



Data Exploration - Univariate Analysis

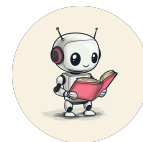


There are a large number of university aged students, and then another peak around middle-age

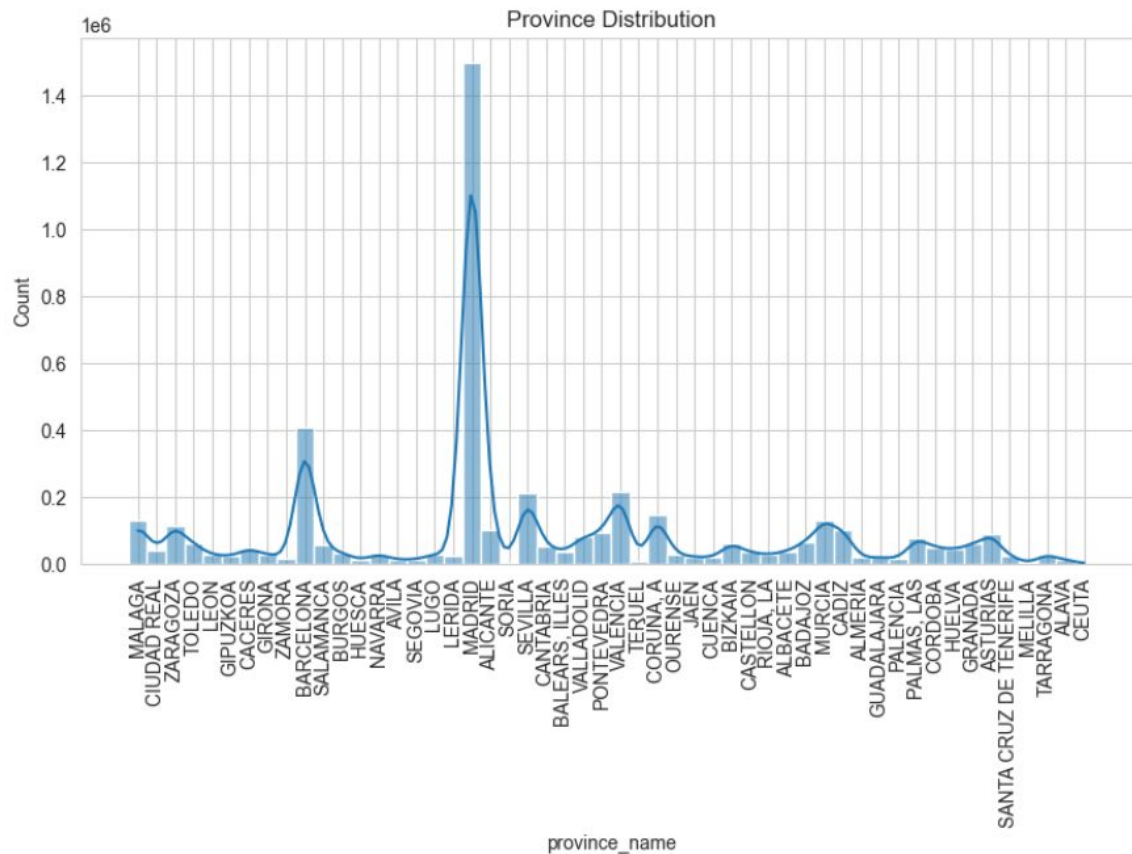


- 👉 Most customers join the bank through the channels **KHE, KAT, KFC, and KHO**.

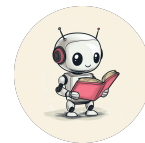
Data Exploration - Univariate Analysis



The most customer from Madrid and Barcelona

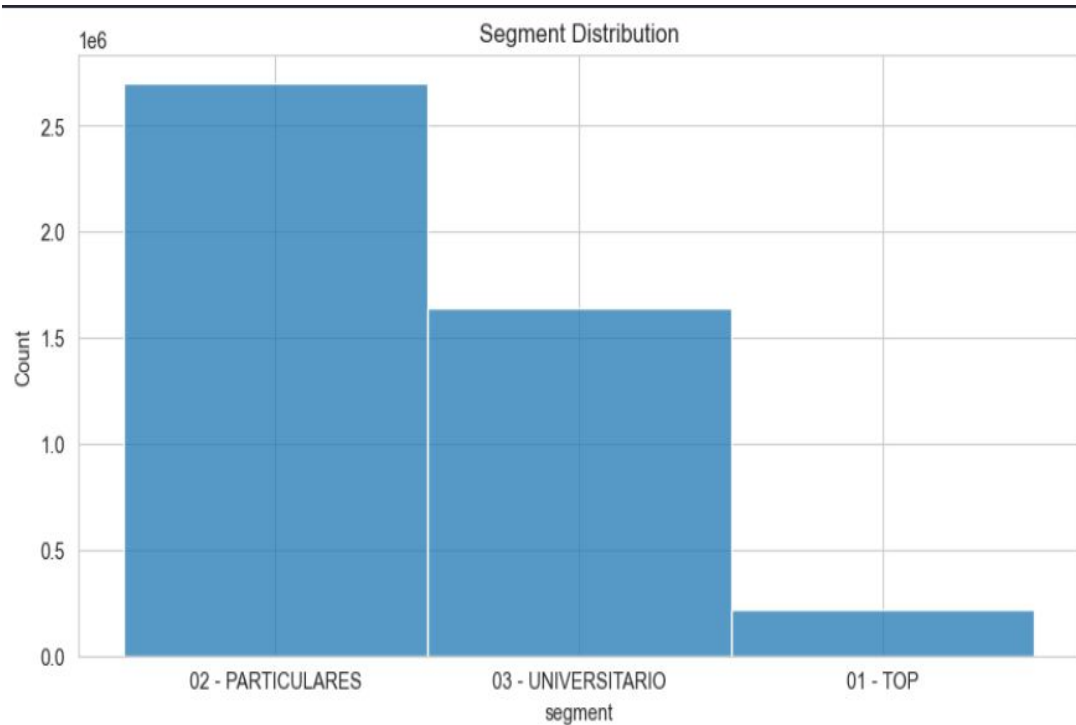


Data Exploration - Univariate Analysis

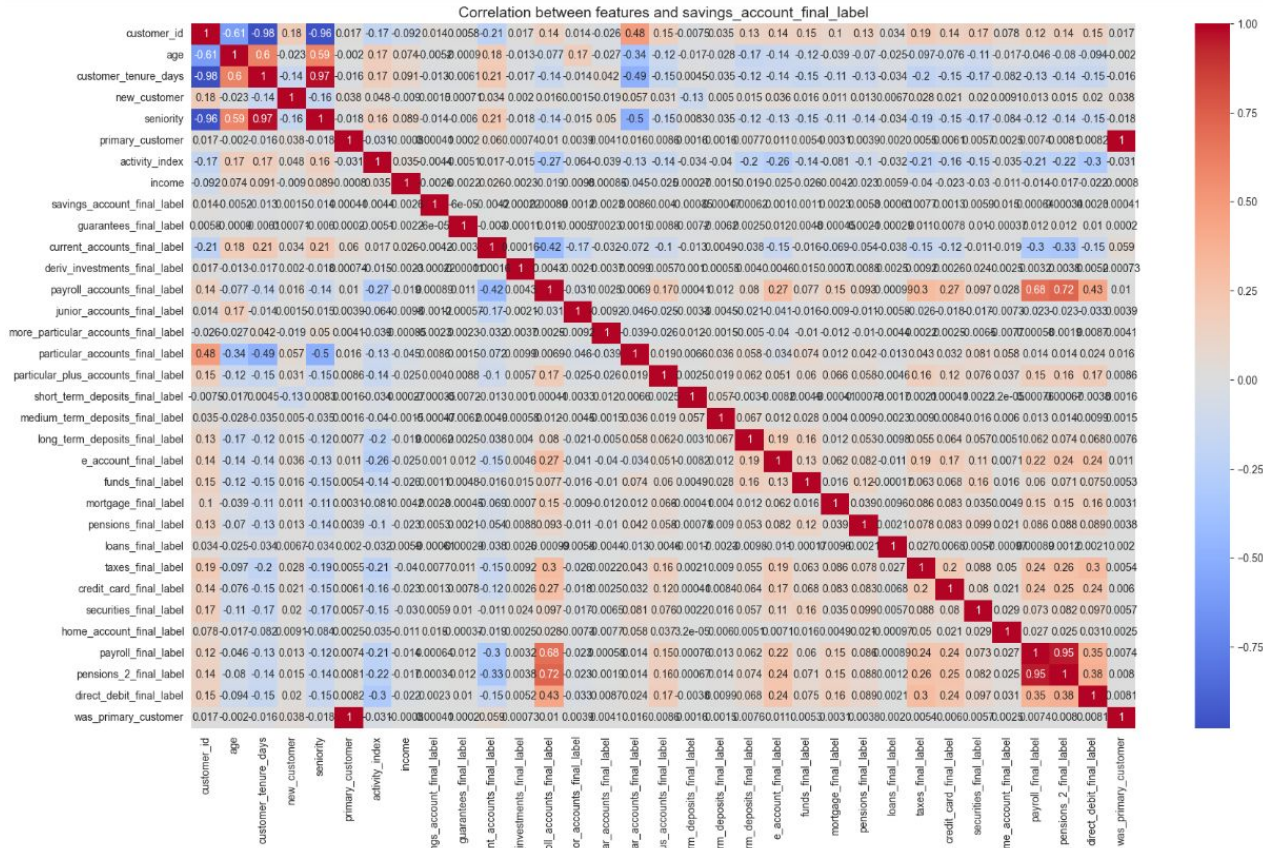


Most customers are individuals and college graduates

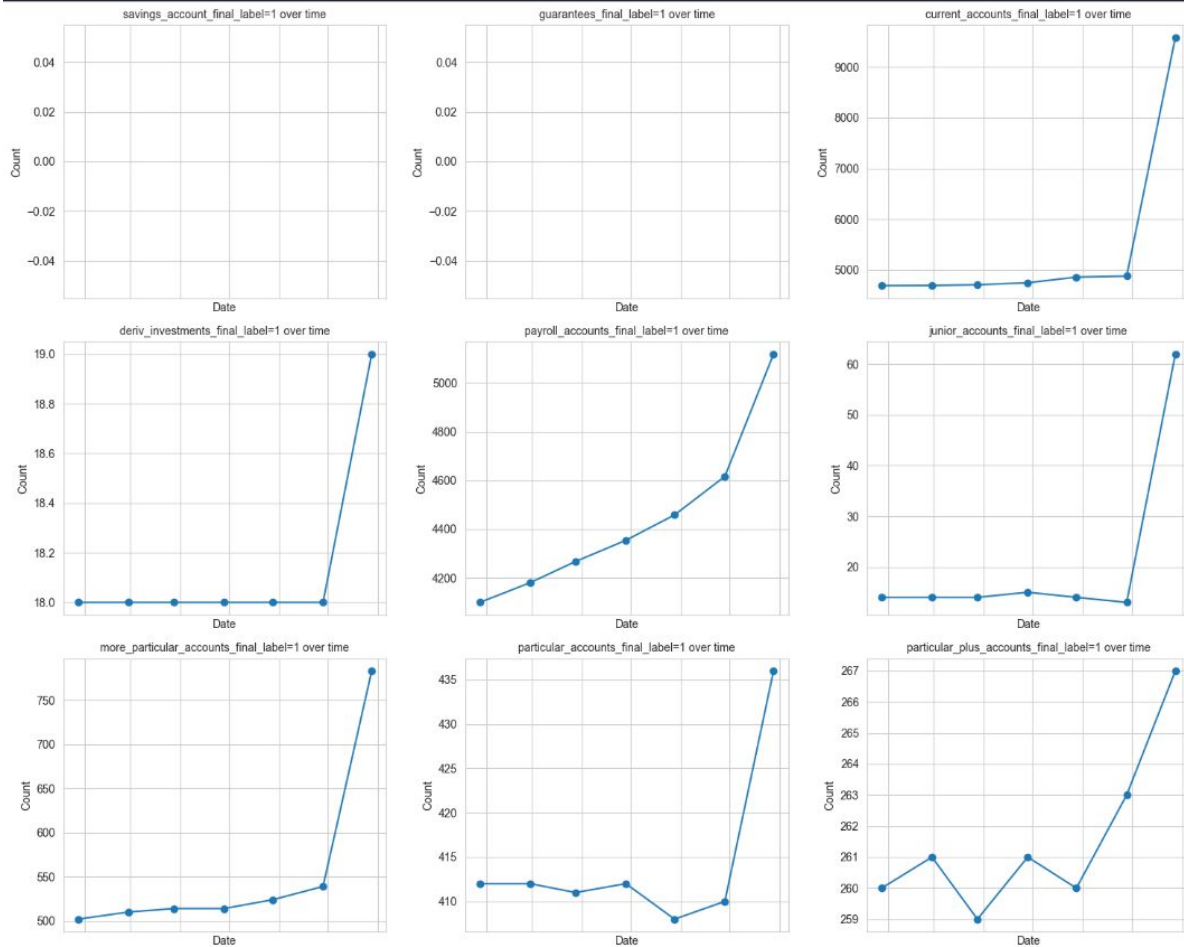
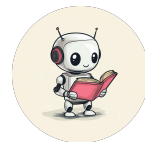
Only a small proportion of customers are VIPs

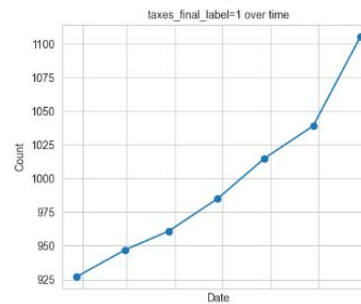
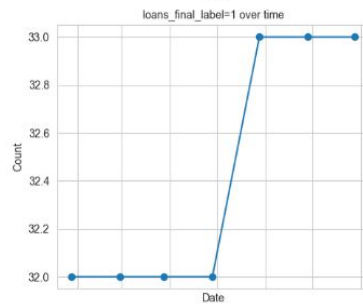
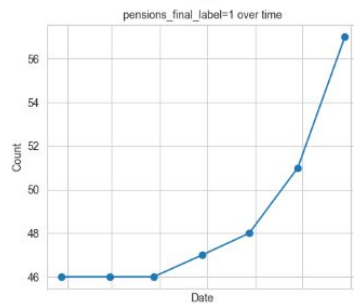
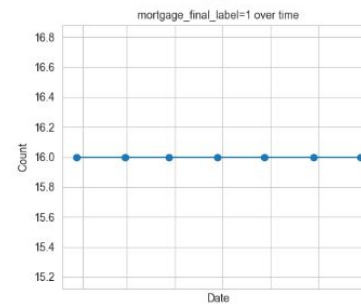
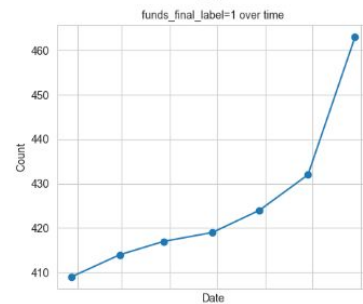
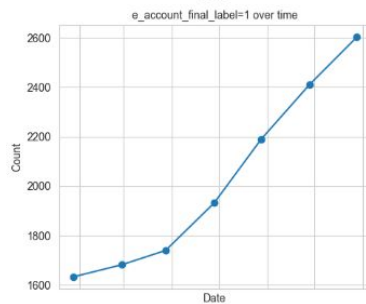
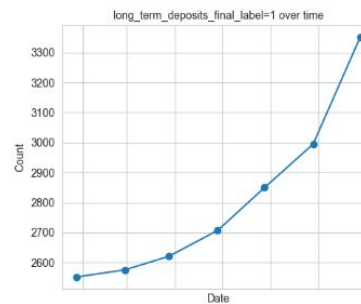
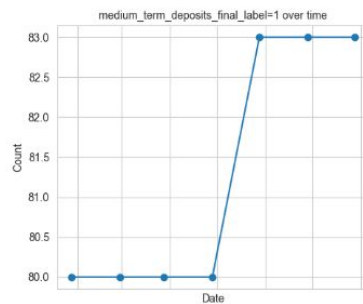
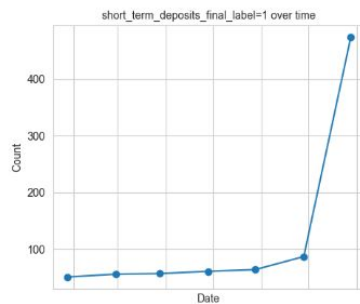
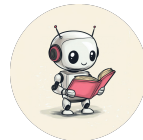


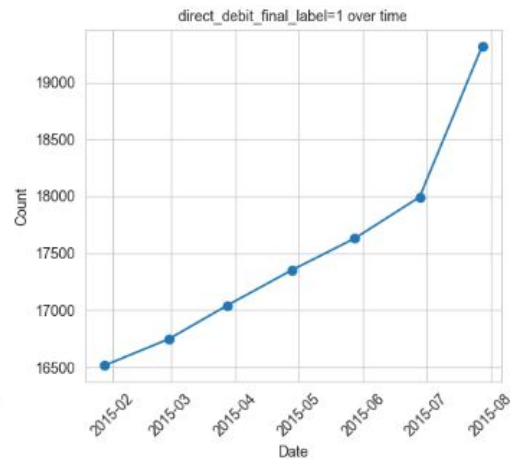
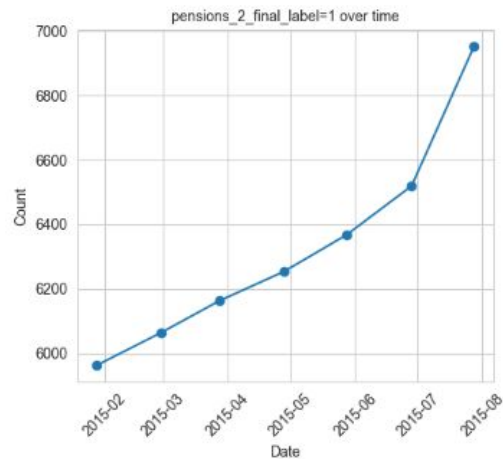
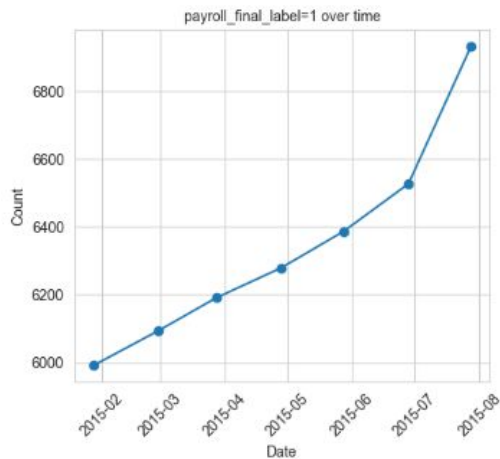
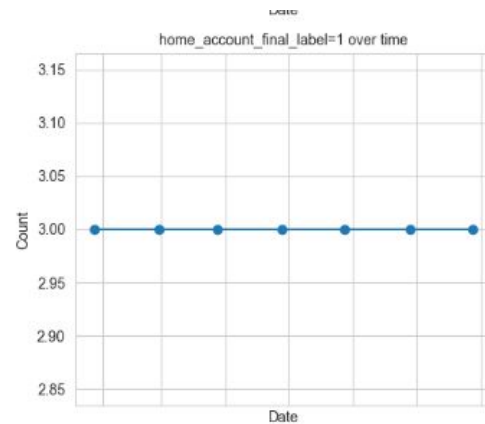
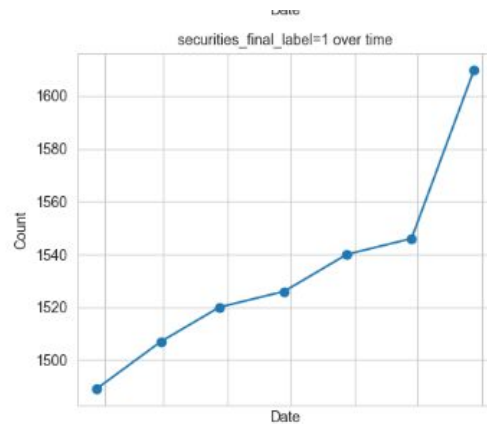
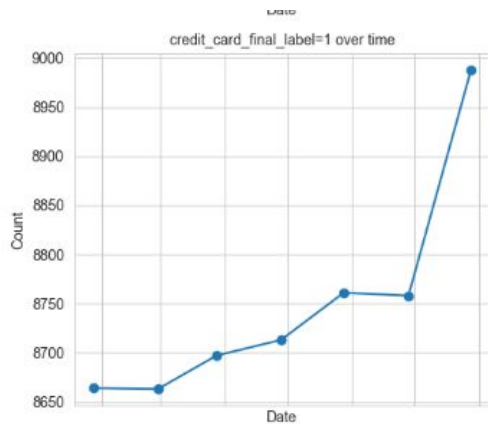
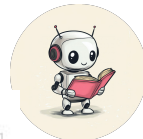
Correlation matrix



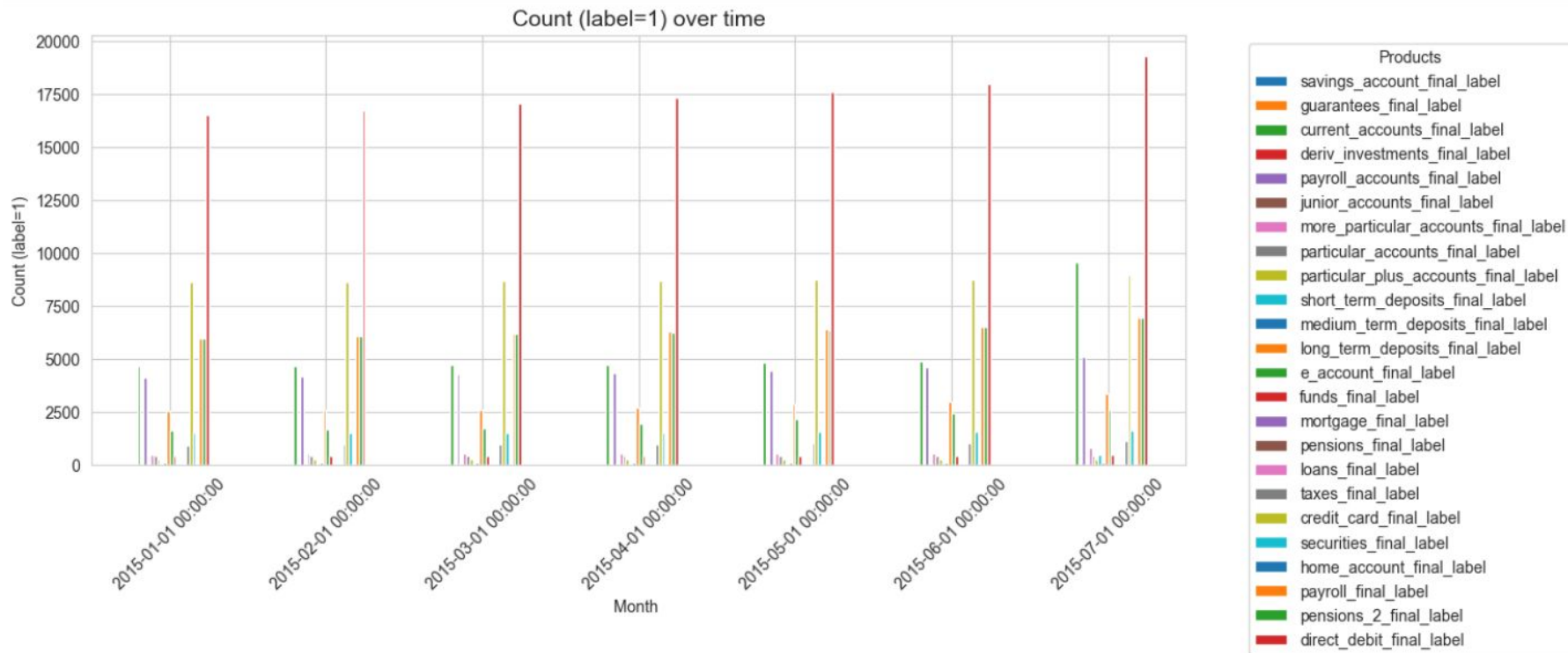
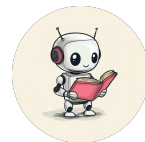
Customer Behaviour over time



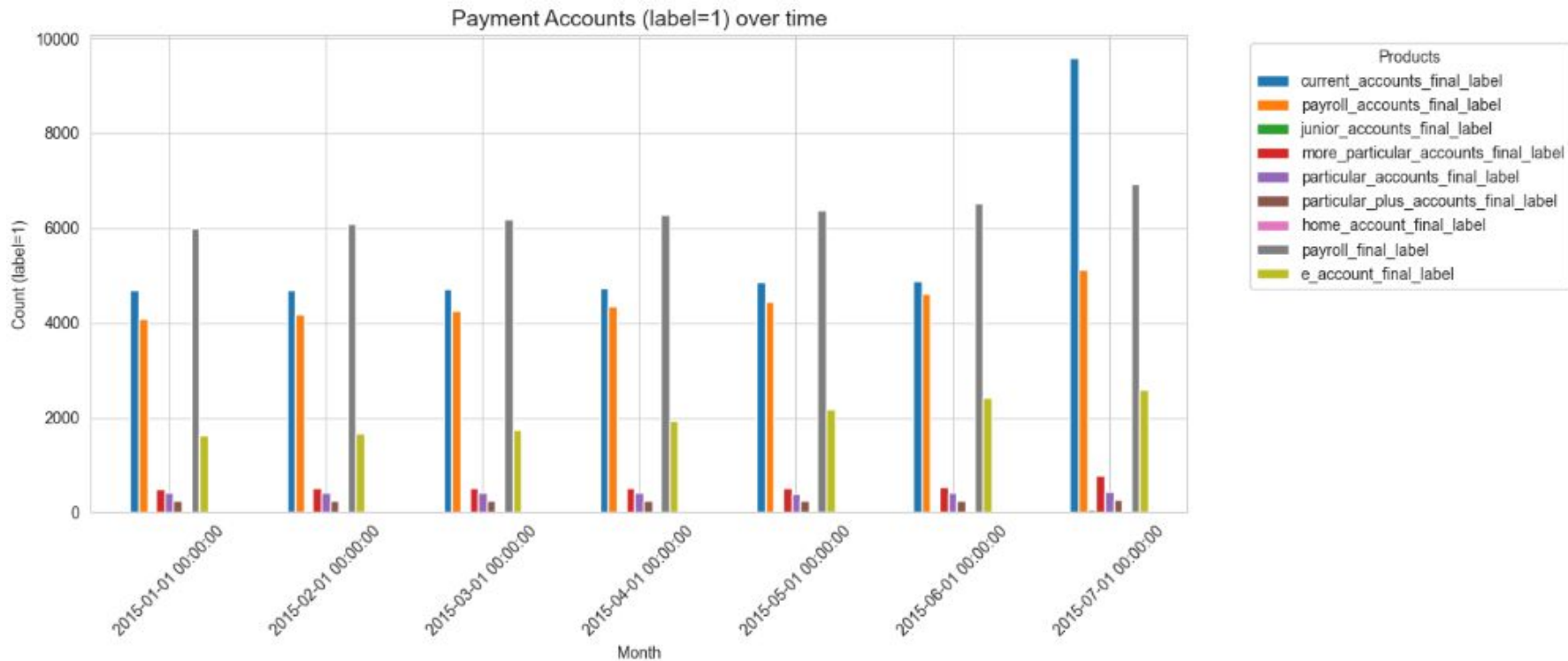
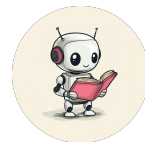




Multivariate Analysis - Customer behaviour



Payments Account Group





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Part 3: Model Development



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Feature Engineering

Feature	Cách tính	Ý nghĩa
<code>year, month, quarter</code>	Tách từ <code>date</code>	Bắt yếu tố thời gian, mùa vụ
<code>day_of_week, is_weekend</code>	<code>date.dayofweek</code> , cờ =1 nếu T7/CN	Nhận diện hành vi khác biệt cuối tuần
<code>is_month_end, is_quarter_end</code>	1 nếu <code>day ≥ 25</code> hoặc <code>month ∈ {3, 6, 9, 12}</code>	Hoạt động mạnh vào cuối kỳ
<code>years_since_registration</code>	<code>customer_tenure_days / 365.25</code>	Thời gian gắn bó với ngân hàng
<code>tenure_category</code>	Phân loại tenure theo bins	Gom nhóm khách hàng mới, lâu năm
<code>age_group</code>	Phân loại tuổi theo khoảng	Phân tích hành vi theo độ tuổi
<code>is_young_adult, is_middle_aged, is_senior</code>	Cờ theo ngưỡng tuổi	Nhận diện nhóm tuổi cụ thể
<code>age_squared</code>	<code>age ** 2</code>	Mô hình hóa quan hệ phi tuyến với tuổi
<code>seniority_years, seniority_category</code>	<code>seniority/12</code> , phân nhóm bins	Mức độ quan hệ theo tháng/năm
<code>is_new_relationship, is_established_relationship</code>	Cờ theo <code>seniority ≤ 6</code> hoặc <code>≥ 24</code>	Khách hàng mới vs. lâu năm

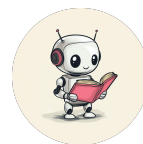


Curious Machine

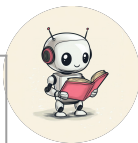
<code>is_employee</code>	<code>1</code> nếu <code>employee_index == 1</code>	Nhận diện nhân viên nội bộ
<code>is_primary_customer_flag</code>	Từ <code>was_primary_customer</code>	Đánh dấu khách hàng chính
<code>is_domestic, is_foreigner</code>	So sánh <code>country_of_residence, foreigner_index</code>	Phân biệt khách trong nước và nước ngoài
<code>age_seniority_interaction, seniority_per_age</code>	Tương tác <code>age × seniority</code>	Khách trẻ nhưng quan hệ lâu có thể rất tiềm năng
<code>age_tenure_ratio</code>	<code>age / (tenure/365 + 1)</code>	Cân bằng tuổi và số ngày gắn bó
<code>channel_type, is_digital_channel</code>	Map <code>channel</code> thành loại, cờ digital	Hành vi giao dịch qua kênh số
<code>is_vip_segment, is_university_segment</code>	Regex từ <code>segment</code>	Nhóm khách hàng đặc biệt
<code>customer_stability_score</code>	<code>(seniority ≥ 12) + (age ≥ 30) + is_primary_customer + is_employee</code>	Thước đo mức độ ổn định
<code>is_stable_customer</code>	<code>1</code> nếu <code>stability_score ≥ 2</code>	Cờ khách hàng trung thành
<code>customer_potential_score</code>	<code>(25 ≤ age ≤ 55) + is_digital_channel + is_domestic</code>	Đo lường tiềm năng tạo giá trị
<code>is_high_potential</code>	<code>1</code> nếu <code>potential_score ≥ 2</code>	Cờ khách hàng có giá trị cao

Metrics for Model Assessment

Standards - Offline Metrics



Tháng 2015	Model	MAP@2	Precision@2	AUC@2	HitRate@2	MAP@3	Precision@3	AUC@3	HitRate@3
Aug	RF	0.0172	0.0177	0.0000	0.0353	0.0187	0.0137	0.0171	0.0400
	XGB	0.0368	0.0205	0.0354	0.0399	0.0377	0.0147	0.0369	0.0413
	LGBM	0.0362	0.0204	0.0345	0.0397	0.0370	0.0146	0.0364	0.0411
Sep	RF	0.0149	0.0155	0.0000	0.0311	0.0164	0.0122	0.0149	0.0353
	XGB	0.0321	0.0181	0.0311	0.0351	0.0330	0.0133	0.0324	0.0367
	LGBM	0.0315	0.0180	0.0301	0.0350	0.0324	0.0132	0.0317	0.0365



Tháng 2015	Model	MAP@2	Precision@2	AUC@2	HitRate@2	MAP@3	Precision@3	AUC@3	HitRate@3
Oct	RF	0.0118	0.0123	0.0000	0.0246	0.0129	0.0097	0.0118	0.0281
	XGB	0.0254	0.0145	0.0245	0.0280	0.0261	0.0106	0.0257	0.0293
	LGBM	0.0245	0.0142	0.0232	0.0275	0.0253	0.0105	0.0247	0.0291
Nov	RF	0.0108	0.0112	0.0000	0.0225	0.0118	0.0088	0.0109	0.0258
	XGB	0.0233	0.0132	0.0225	0.0257	0.0239	0.0096	0.0236	0.0270
	LGBM	0.0223	0.0129	0.0210	0.0253	0.0230	0.0095	0.0226	0.0267
Dec	RF	0.0090	0.0093	0.0000	0.0186	0.0096	0.0070	0.0091	0.0205
	XGB	0.0190	0.0106	0.0186	0.0206	0.0194	0.0076	0.0192	0.0213
	LGBM	0.0181	0.0103	0.0174	0.0201	0.0186	0.0075	0.0184	0.0211

TUNING MODEL



2015	Model	MAP@2	Precision@2	AUC@2	HitRate@2	MAP@3	Precision@3	AUC@3	HitRate@3
Aug	RF	0.0172	0.0177	0.0000	0.0353	0.0187	0.0137	0.0171	0.0401
	XGB	0.0368	0.0205	0.0354	0.0396	0.0376	0.0147	0.0369	0.0413
	LGBM	0.0368	0.0205	0.0353	0.0399	0.0376	0.0147	0.0370	0.0413
Sep	RF	0.0149	0.0155	0.0000	0.0311	0.0164	0.0122	0.0149	0.0354
	XGB	0.0315	0.0179	0.0304	0.0347	0.0325	0.0132	0.0318	0.0365
	LGBM	0.0319	0.0181	0.0308	0.0351	0.0328	0.0132	0.0322	0.0365
Oct	RF	0.0118	0.0123	0.0000	0.0246	0.0129	0.0097	0.0118	0.0281
	XGB	0.0244	0.0141	0.0234	0.0272	0.0253	0.0105	0.0247	0.0291
	LGBM	0.0251	0.0144	0.0243	0.0278	0.0259	0.0105	0.0254	0.0291



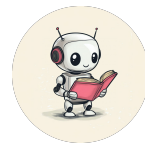
2015	Model	MAP@2	Precision@2	AUC@2	HitRate@2	MAP@3	Precision@3	AUC@3	HitRate@3
Nov	RF	0.0108	0.0112	0.0000	0.0225	0.0119	0.0088	0.0109	0.0258
	XGB	0.0224	0.0129	0.0213	0.0251	0.0232	0.0095	0.0228	0.0268
	LGBM	0.0230	0.0131	0.0222	0.0255	0.0237	0.0096	0.0233	0.0267
Dec	RF	0.0090	0.0093	0.0000	0.0187	0.0096	0.0070	0.0091	0.0205
	XGB	0.0182	0.0103	0.0175	0.0201	0.0188	0.0076	0.0185	0.0213
	LGBM	0.0187	0.0104	0.0184	0.0203	0.0192	0.0075	0.0190	0.0211



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Machine

Part 4: Deployment

Deployment

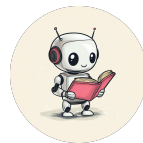




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Part 5: Action

Action



Which department will the model results be sent to?

- ✓ Marketing Department.
- ✓ Sales/Branch Operations.
- ✓ Customer Relationship Management (CRM) Team.
- ✓ Product Strategy Department.

How to operate the results from the model?

Scenario:

- Each month, when customer transaction data is updated, the LightGBM model predicts the top-3 products each customer is most likely to adopt.
- Alternatively, when a customer logs into the mobile/online banking app, the system calls the API to generate real-time recommendations.

Input:

- Customer profile data (age, gender, income segment, existing products, transaction behavior, digital engagement).

Output:

- For each customer: **Top-3 recommended products**, each with an associated **probability score** (e.g., Credit Card 0.35, Savings Account 0.28, Insurance 0.22).



Q & A